

UNIVERSITY OF DAR ES SALAAM



**COLLEGE OF INFORMATION AND COMMUNICATION TECHNOLOGIES
(CoICT)**

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (CSE)

END-SEMESTER PROJECT REPORT

PROJECT TITLE: UDSM CONNECTED INTELLIGENT SYSTEM.

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LIST OF ABBREVIATIONS

UDSM	University Of Dar-es-salaam
COICT	College Of Information and Communication Technology
AI	Artificial Intelligence
API	Application Programming Interface
UDSM	University of Dar es Salaam
UAT	User Acceptance Testing
UCIS	UDSM Connected Intelligent System
SSO	Single Sign-On
ARIS	Academic Registration Information System
LMS	Learning Management System
FYP	Final Year Project
PTMS	Postgraduate Tracker Management System
UC	Use case

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1. CHAPTER ONE

1.1. GENERAL INTRODUCTION

The rapid advancement of information and communication technology (ICT) has significantly transformed the operations of higher learning institutions worldwide. Universities increasingly rely on digital systems to support academic activities, administrative processes, communication, and student services. Efficient information systems play a crucial role in enhancing service delivery, improving accessibility to information, and supporting informed decision-making within the university environment.

At the University of Dar es Salaam (UDSM), several academic and administrative digital systems are already in place, including academic registration platforms, e-learning systems, and communication channels. However, these systems operate independently, and information is often disseminated through multiple platforms such as noticeboards, emails, websites, and informal messaging groups. This fragmentation creates challenges for students and staff in accessing timely and accurate information, navigating campus facilities, and responding effectively during emergencies.

This project proposes the development of a centralized digital university that will provide different university services, such as announcements, news, a campus map, and a gateway to academic systems such as ARIS and LMS. The platform aims to improve communication, simplify access to academic services, support campus navigation, and strengthen emergency response mechanisms. By consolidating essential university services into a single, trusted platform. The system aims to enhance efficiency, usability, and the overall campus experience for students and staff.

1.2. STATEMENT OF THE PROBLEM

Despite the availability of various academic and administrative systems at UDSM, access to essential information and services remains fragmented. Important announcements are distributed across noticeboards, social media platforms, and different offices, causing many students and staff to miss critical updates.

The specific objectives of the project are:

- To collect and analyze the user requirements.
- To design the system architecture, interfaces, and database designs.
- To develop the mobile application.
- To test the system for functionality, usability, and reliability.

1.4. SIGNIFICANCE OF THE PROJECT

The proposed system is expected to benefit various stakeholders within the UDSM community.

Students will have easy access to academic platforms, timely information, campus navigation tools, and support resources, thereby enhancing their academic experience and overall well-being. Staff and university administrators will benefit from improved communication channels, reduced workload through automation, and better engagement with students.

The university as an institution will benefit from improved operational efficiency, better dissemination of official information, enhanced safety measures, and stronger connections between students, staff, and alumni. Additionally, the system provides a scalable framework that can support future digital transformation initiatives at UDSM.

1.5. PROJECT SCOPE

The scope of this project focuses on the design and development of a centralized digital platform that provides unified access to key university services. The system will include centralized announcements, access to academic platforms, a campus map and navigation feature, and emergency contact access.

Additional features within the scope include a personalized user dashboard, an AI-based virtual assistant, campus life and events management, mentorship and alumni connection, and digital feedback mechanisms.

The project will focus on system design, functionality, and usability rather than replacing existing academic systems. Development will be limited to the UDSM environment.

2. CHAPTER TWO: LITERATURE REVIEW

2.1. OVERVIEW OF EXISTING SYSTEMS

Currently, students, staff, and visitors at UDSM rely on a combination of Google Maps, printed maps, noticeboards, and personal assistance to navigate the campus.

Although Google Maps provides a general overview of the university, it does not contain accurate or up-to-date information about many buildings, lecture halls, and department offices. Some buildings are mislabeled, missing, or have changed names, making it difficult for users to find their way around campus. As a result, many students and visitors still depend on asking others for directions.

For communication, the university primarily uses WhatsApp groups, noticeboards, the official UDSM website, and social media platforms such as Instagram and Facebook. While these platforms are helpful, they are not well integrated. Students may miss updates if they are not part of the right WhatsApp groups or if they do not regularly check noticeboards or social media pages.

In emergencies, students and staff usually rely on phone calls, word-of-mouth, or physically visiting university offices. Although some emergency contact numbers are shared on posters or WhatsApp groups, they are not centrally organized or easily accessible in one place. The lack of a unified emergency system can cause delays when immediate help is needed.

2.2. RELATED WORKS

Many universities worldwide have developed mobile applications to improve student experience, campus navigation, and access to university services. These systems provide valuable lessons for the development of the proposed UDSM Mobile Application.

A notable example is the University of Notre Dame mobile application. This app allows students and staff to receive important announcements, access academic calendars, and find emergency contact information easily. It also includes an interactive campus map that provides step-by-step directions to lecture halls, offices, and dining areas. Additionally, the app integrates with academic systems, enabling students to check their schedules and grades in one place.

Similarly, the University of Houston has implemented a centralized mobile application that serves as a digital hub for university information. The app delivers push notifications for important updates, features an events calendar, and includes safety tools for reporting security concerns and receiving emergency alerts. This aligns closely with UDSM's goal of improving emergency communication and response.

2.3. IDENTIFIED GAPS

Despite these advancements in other institutions, UDSM still faces significant gaps in its digital infrastructure.

1. One major gap is the absence of an accurate campus navigation system. Google Maps is not sufficient for detailed university mapping, making it difficult for students, especially newcomers, to locate key facilities.
2. Another gap is the lack of a reliable and unified communication platform. Current methods, such as WhatsApp groups and noticeboards, are inconsistent and exclude some students and staff, leading to missed information and confusion.
3. Furthermore, there is no centralized emergency contact system that allows users to quickly reach security or medical services. In critical situations, this delay can be dangerous.

These challenges clearly justify the need for a single, integrated UDSM Mobile Application that combines campus navigation, real-time announcements, and emergency support in one platform. Implementing such a system would significantly improve communication, accessibility, and overall student experience at the University of Dar es Salaam.

3. CHAPTER THREE: METHODOLOGY

3.1. INTRODUCTION

To implement the UDSM connected intelligent system, we are expecting a close collaboration with the UDSM students and lecturers to make sure that, as the development goes on, there is a meeting and a wide coverage of the system's requirements. Our end goal is to have a fully working system that is scalable, as more components may be required to be added to it.

3.2. DEVELOPMENT METHODOLOGY

The main methodology that we will implement is the Agile-Scrum methodology due to its iterative and adaptable nature, allowing us to respond to evolving user needs and gather and implement feedback, also allowing users to experience the application in the process of its making.

Scrum Framework:

The focus is on dividing the development process into smaller, manageable units called sprints and ensuring user feedback is collected and acted upon

The Scrum framework is a lightweight and adaptive methodology used to address complex problems and deliver high-value products iteratively. Scrum, rooted in Agile principles, organizes work into time-boxed iterations called sprints, typically lasting two to four weeks. In this framework, we are expecting a quicker way to implement the development since we are required to implement both the mobile application and a chatbot that can be integrated into the UDSM website.



Figure 2: Agile Methodology.

3.3. REQUIREMENT ANALYSIS

To gather the requirements for the UDSM connected intelligent initiative, the main requirement gathering method is interviewing. Kothari (2004) says that an interview is a powerful exchange of ideas, the answering of questions, and communication between two or more persons.

Overall, the following techniques are employed:

1. Interviews: Direct consultations with stakeholders (including the UDSM staff and the students, who are the major players in the making) to understand their needs, challenges, and expectations.
2. Stakeholder Consultations: Meetings with UDSM lecturers and students to refine and validate the initial requirements, ensuring alignment with institutional goals.
3. Questionnaires: Using Google Forms, passed on to student groups of all years, as well as face-to-face interrogation with students of different years

3.4. FINDINGS FROM THE REQUIREMENTS GATHERING PROCESS.

To ensure that the proposed UDSM Mobile Application responds effectively to real needs within the university, a requirements-gathering process was conducted. The purpose of this process was to understand the actual challenges faced by students, staff, and alumni in their daily interactions with university services, communication channels, and campus facilities. The team used interviews and personal observations as primary methods of data collection.

As final-year students at the University of Dar es Salaam, the project team has personally experienced many of the communication and accessibility challenges identified during the interviews.

One major issue is the reliance on informal communication channels for important academic announcements. There have been several cases where critical updates, such as class cancellations, changes in lecture venues, or revisions to the examination timetable, did not reach all students in time. This has led to confusion, missed deadlines, and, in some cases, students missing examinations. These problems could have been avoided if there had been a reliable, centralized platform that sent real-time notifications to all students.

Additionally, access to emergency services on campus is not well organized. Students and staff often struggle to quickly find contact details for auxiliary police, the University Hospital, or student support offices during urgent situations. This delay can be risky, especially in medical or security-related emergencies.

Based on these findings, it is clear that UDSM urgently needs a structured digital platform that provides centralized announcements, an interactive campus navigation system, and fast access to emergency services through real-time notifications.

3.5. IMPLEMENTATION OF THE UCIS.

The mobile application will be developed primarily for the Android platform due to its widespread use among UDSM students and staff. The interface will be simple, intuitive, and optimized for low data consumption.

The backend will manage authentication, user roles, data processing, notifications, and integration with academic systems. Secure APIs will be used to communicate with the mobile application.

An AI-powered virtual assistant will be integrated through an external API. The assistant will respond to common queries about academics, campus navigation, deadlines, and procedures. It will also guide users to appropriate university services.

The database will store user profiles, announcements, events, booking records, feedback, and logs of chatbot interactions. Data security measures such as access control and encryption will be implemented.

3.6. TESTING AND QUALITY ASSURANCE

Testing will be conducted throughout the development lifecycle. Unit testing verifies individual modules, integration testing ensures components work together, and system testing validates the complete system.

User Acceptance Testing (UAT) will be conducted with selected students and staff to ensure the system meets user expectations and functional requirements.

Quality assurance will be done to ensure the whole system works correctly, safely, quickly, and is easy to use. The AI assistant works well and can be trusted, gives accurate answers, and understands common student/user questions.

3.7. DEPLOYMENT

The system will be set up in the cloud, which means it will be accessible from anywhere and can handle a large number of users at the same time. The backend, which manages data, and the database where information is stored, will be kept safe and secure.

Deployment will involve final testing, server configuration, database setup, API integration, and the release of the application. Monitoring tools will be used to ensure system stability.

Post-deployment maintenance will include bug fixes, feature updates, performance optimization, and security updates.

4. CHAPTER FOUR: SYSTEM ANALYSIS AND DESIGN

4.1. STAKEHOLDER IDENTIFICATION

The next phase of the project involved gathering requirements from the stakeholders who, in one way or another, are meant to interact and benefit from the application. Before initiating this process, those key stakeholders had to be identified first so that it would be easy to determine whom to interview to collect the requirements. These include students, staff, alumni, visitors, and administrators. Each of these groups has different expectations from the system, thus making it important to gather their requirements efficiently.

4.2. REQUIREMENTS GATHERING

Requirement Category

- a) **Mandatory:** These are core requirements that the system must include. They are necessary for the application to fulfil its main purpose and therefore can't be omitted or changed.
- b) **Optional:** These are additional features that improve the system but are not essential for its operation. The application can still function properly without them, and they may be added in future updates.
- c) **Measurable:** These requirements are defined using specific and quantifiable criteria. They can be objectively verified using clear metrics or numerical values.
- d) **Evident:** These are simple and clearly defined requirements that are easy for all stakeholders to understand and require no further explanation.

4.2.1 USER REQUIREMENTS

Ref No.	DESCRIPTION		CATEGORY
UR1	Authentication & Access		
	UR1.1	As a student, I want to log in using my UDSM mail credentials via Single Sign-On (SSO).	Mandatory

	UR1.2	As a user, I want to view a personalized dashboard displaying relevant updates.	Evident
	UR1.3	As a user, I want to create and manage my profile within the app.	Evident
	UR1.4	As a new student, I want the chatbot to help me understand how to use university services.	Evident
UR2	Campus Navigation & Location Services		
	UR2.1	As a user, I want to search for locations based on categories.	Evident
	UR2.3	As a user, I want to ask the chatbot where a place is located on campus (e.g., Library, lecture hall, cafeteria).	Mandatory
UR3	News, Announcements, & Events		
	UR3.1	As a student, I want to receive news updates about ongoing academic, social, and extracurricular activities.	Evident
	UR3.2	As a user, I want to filter news based on category (e.g., academics, sports, cultural events).	Optional
	UR3.3	As a user, I want to ask the chatbot latest university news.	Mandatory

	UR3.4	As a student, I want to access workshop schedules, debates, arts, and sports events directly from the app.	Optional
	UR3.5	As a user, I want to view the university almanac displaying upcoming events in a calendar format.	Evident
	UR3.6	As a user, I want to filter announcements based on my department or role.	Optional

	UR3.7	As an admin, I want to be able to post news updates and announcements for different user groups (students, staff, alumni).	Mandatory
	UR3.8	As an admin, I want to be able to update the university almanac calendar.	Evident
UR4	Emergency & Student Support Services		
	UR4.1	As a student, I want to access emergency contacts for the university hospital, auxiliary police, dean of students, accommodation bureau, and counseling unit.	Mandatory
	UR4.2	As an admin, I want to be able to update emergency contact numbers as needed.	Mandatory
	UR4.3	As a user, I want the chatbot to guide me to the nearest emergency service location.	Mandatory
UR5	Others		
	UR5.1	As an alumnus, I want to connect with fellow alumni for networking and career opportunities.	Mandatory
	UR5.2	As an alumnus, I want to receive notifications about alumni reunions and events.	Evident
	UR5.3	As a user, I want to access news updates about university projects, graduations, and achievements.	Evident
	UR5.4	As an alumnus, I want to be informed about job vacancies and internship opportunities.	Optional

	UR5.5	As an alumni, I want to participate in an alumni mentorship program, allowing me to mentor current students.	Optional
	UR5.6	As a staff, I want to access teaching schedules and meeting alerts.	Mandatory
	UR5.7	As an admin, I want to manage user roles and permissions.	Mandatory
	UR5.8	As an admin, I want to be able to allow multiple administrators, enabling different faculty admins to manage their respective departments.	Evident
	UR5.9	As an admin, I want to be able to generate reports on user engagement, app usage, and feedback.	Measurable

4.2.2 FUNCTIONAL REQUIREMENTS

Ref. No.	REQUIREMENT DESCRIPTION		CATEGORY
FR1	Authentication & User Management		
	FR1.1	The system shall authenticate users via Single Sign-On (SSO) using UDSM Mail credentials.	Mandatory
	FR1.2	The system shall allow role-based access control (RBAC) to restrict sensitive data access.	Mandatory
	FR1.3	The system shall allow students, staff, and alumni to edit their profiles and update personal information.	Evident

	FR1.4	The system shall allow administrators to manage user roles, permissions, and accounts.	Mandatory
	FR1.5	The chatbot shall send event reminders before the scheduled time.	Optional
FR2	Campus Navigation & Maps		
	FR2.1	The system shall provide an interactive campus map with location-based search functionality.	Measurable
	FR2.2	The system shall allow users to search locations by name or category (e.g., lecture halls, cafeterias, libraries).	Evident
	FR2.3	The system shall allow administrators to update location data for accuracy.	Mandatory
	FR2.4	The chatbot shall provide direct navigation to the event location via map integration.	Mandatory

	FR2.5	The system shall allow real-time updates to location data when new buildings or services are introduced.	Mandatory
FR3	Announcements & Notifications		
	FR3.1	The system shall allow administrators to post announcements targeted at specific user groups.	Mandatory
	FR3.2	The system shall send push notifications for important updates, including announcements from the administration and student government.	Mandatory
	FR3.3	The chatbot shall send event reminders before scheduled time.	Optional

	FR3.4	The system shall allow students to customize notification preferences to filter relevant updates and prevent information overload.	Optional
	FR3.5	The system shall automatically remove outdated announcements after their expiration date.	Evident
FR4	University Almanac & Events		
	FR4.1	The system shall display the university almanac in a calendar format.	Evident
	FR4.2	The system shall allow administrators to add, update, or remove events in the university almanac.	Mandatory

	FR4.3	As a user, I want to ask the chatbot about upcoming events.	Mandatory
	FR4.4	The system shall allow users to view event details (e.g., date, time, location, description).	Evident
	FR4.5	The system shall allow users to subscribe to specific events for reminders.	Optional
FR5	News & Updates		
	FR5.1	The system shall allow administrators to post news articles under different categories (academic, sports, cultural, alumni, university projects).	Mandatory
	FR5.2	The system shall allow users to filter news based on categories.	Optional

	FR5.3	The system shall archive old news articles but allow users to search and retrieve past news items.	Optional
FR6	Emergency Contact System		
	FR6.1	The system shall provide a dedicated emergency contacts page accessible from the main menu.	Mandatory
	FR6.2	The system shall allow administrators to update emergency contact numbers for key services (hospital, police, counseling unit, student accommodation bureau, etc.).	Mandatory
	FR6.3	The system shall enable a quick-call feature allowing users to call emergency contacts directly from the app.	Optional
	FR6.4	As a user, I want the chatbot to guide me to the nearest emergencies (accident, illness, theft , fire).	Mandatory
FR7	Alumni Features		
	FR7.1	The system shall allow alumni to register and update their profiles separately from active students.	Mandatory
	FR7.2	The system shall send event notifications for alumni reunions and networking events.	Evident
	FR7.3	The system shall provide a job board for alumni to access job postings and internship opportunities.	Optional
	FR7.4	The system shall enable an alumni networking feature to connect alumni based on industry, graduation year, or college.	Optional
FR8	System Administration & Content Management		

	FR8.1	The system shall allow multiple system administrators to manage content based on their assigned faculties or departments.	Mandatory
	FR8.2	The system shall allow administrators to schedule announcements and news articles for future posting.	Optional
	FR8.3	The system shall provide a feedback mechanism for users to report app issues or suggest improvements.	Optional
FR9	Unified Dashboard and Academic System Integration		
	FR9.1	The system shall provide a unified dashboard displaying announcements, deadlines, academic alerts, and quick links to integrated university systems.	Mandatory
	FR9.2	The dashboard shall personalize displayed information based on user roles (student, staff, alumni, admin).	Mandatory
	FR9.3	The system shall provide direct and secure access to existing academic systems, including LMS, ARIS, FYP Portal, and PTMS portal	Mandatory
FR10	Campus Life and Mentorship Features		
	FR10.1	The system shall display information about student organizations and extracurricular opportunities available on campus	Optional
	FR9.2	The system shall enable students to connect with alumni and academic staff for mentorship, career guidance, and professional support.	Optional

4.2.3 NON-FUNCTIONAL REQUIREMENTS

Ref. No.	REQUIREMENT DESCRIPTION		CATEGORY
NFR1	Operational Requirements		
	NFR1.1	The application shall be available on Android (Google Play Store) and iOS (Apple App Store) to ensure accessibility for all stakeholders.	Mandatory
	NFR2.1	The application shall support 10,000 concurrent users without performance degradation.	Measurable
	NFR2.2	The dashboard shall load within 3 seconds under normal conditions.	Measurable
	NFR2.3	The chatbot shall process search queries (e.g., finding a lecture hall) within 2 seconds.	Measurable
	NFR2.4	Push notifications shall be delivered to users within 5 seconds of an announcement being published.	Measurable
	NFR2.5	System response time shall not exceed 1 second for navigation within the app.	Measurable
	NFR2.6	The maps interface shall support zooming, panning, and searching for locations.	Evident
NFR3	Security Requirements		
	NFR3.1	The system shall use Single Sign-On (SSO) authentication via UDSM Mail to ensure only verified users access the app	Mandatory
	NFR3.2	All data transmissions shall be encrypted using HTTPS/ TLS 1.2 or higher to protect user data.	Mandatory

	NFR3.3	The app shall enforce role-based access control (RBAC) to restrict unauthorized access to sensitive information.	Mandatory
	NFR3.4	The system shall log all administrative activities for security auditing.	Mandatory
	NFR3.5	Users shall be automatically logged out after 15 minutes of inactivity to prevent unauthorized access.	Mandatory
	NFR3.6	The system shall have two-factor authentication (2FA) for admin users to enhance security.	Optional
	NFR3.7	Emergency contact numbers shall be restricted from external access, ensuring only registered users can view them.	Mandatory
	NFR3.8	Users shall be notified of any unauthorized login attempts to their accounts.	Optional
	NFR3.9	The system shall automatically detect and block suspicious activities, such as multiple failed login attempts.	Mandatory
NFR4	Cultural & Political Requirements		
	NFR4.1	The system shall comply with Tanzanian data protection laws, ensuring user privacy is protected.	Mandatory
	NFR4.2	The app shall adhere to University of Dar es Salaam's branding guidelines (colors, fonts, logos, and visual identity).	Mandatory

	NFR4.3	The app shall ensure accessibility compliance with WCAG 2.1 Level AA standards, supporting students with disabilities.	Mandatory
	NFR4.4	Student government (DARUSO) announcements shall be separate from university administration announcements to avoid confusion.	Evident
	NFR4.5	Any changes to the app's terms and conditions and all updates shall be communicated to users via push notifications.	Mandatory

4.3. USECASE ANALYSIS

4.3.1 OVERVIEW

The **UDSM Connected Intelligent System** is a centralized platform integrating multiple university services, academic systems, and campus functionalities. It enables students, staff, alumni, and visitors to access personalized dashboards, interact with AI assistance, navigate campus, and connect with alumni networks. The system ensures secure authentication via SSO and role-based access management.

4.3.2 ACTORS

Actor	Description
User	Student, staff, or alumni using the system to access dashboards, AI assistant, academic systems, campus services, or mentorship programs.
Admin	Manages system operations, posts announcements, manages user roles, and monitors performance.
Visitor	External users (prospective students, guests) can access limited features such as the AI assistant and campus navigation.
Alumni	Graduates accessing alumni-specific features, including login, registration, and networking.
External Academic Systems (ARIS, LMS, FYP, PTMS)	Provide integrated academic data and services to the system.
AI Assistant API	Processes user queries and provides intelligent guidance.
UDSM Campus Map API	Provides real-time campus navigation and location services.

4.3.3 USECASES

i. Login via Single Sign-On (SSO)

Field	Description
Use Case ID	UC-01
Use Case Name	Login via Single Sign-On (SSO)
Actors	User
Preconditions	User has valid UDSM credentials; User has internet access and system interface
Basic Flow	1. User opens system interface 2. Clicks Login 3. Redirected to SSO page 4. Enters credentials 5. System validates with SSO 6. Access granted to Unified Dashboard
Alternative Flows	• Invalid credentials → "Invalid credentials" message • SSO unavailable → "Authentication service unavailable" message
Postconditions	User is authenticated and accesses personalized dashboard
Exceptions	1. System timeout during authentication. 2. Network connection lost during login process. 3. User account suspended or deactivated.

ii. Access Unified Dashboard

Field	Description
Use Case ID	UC-02
Use Case Name	Access Unified Dashboard
Actors	User
Preconditions	User is logged in via SSO
Basic Flow	1. User selects Dashboard 2. System displays announcements, academic alerts, deadlines, quick

	links
Alternative Flows	Dashboard fails to load → error message displayed
Postconditions	User can view and interact with dashboard elements
Exceptions	1. Session expired while accessing dashboard. 2. Database connection failure. 3. Unauthorized access attempt.

iii. Interact with AI Virtual Assistant

Field	Description
Use Case ID	UC-03
Use Case Name	Interact with AI Virtual Assistant
Actors	User, Visitor
Preconditions	User/Visitor can access system interface
Basic Flow	1. Select AI Assistant 2. System connects to AI API 3. Submit query/select option 4. System responds with guidance
Alternative Flows	• AI API unavailable → error message • Invalid query → prompt to rephrase
Postconditions	User receives relevant guidance or instructions
Exceptions	1. AI Assistant API returns malformed or empty response. 2. API request timeout. 3. System unable to process user input format.

iv. Access Academic Systems (LMS, ARIS, FYP, PTMS)

Field	Description
Use Case ID	UC-04
Use Case Name	Access Academic Systems
Actors	User
Preconditions	User is logged in and authorized
Basic Flow	1. Select Academic Systems 2. Display LMS, ARIS, FYP, PTMS options 3. Select system 4. Redirect to selected system
Alternative Flows	External system unavailable → error message
Postconditions	User accesses selected academic system successfully
Exceptions	1. User not authorized for selected academic system. 2. Redirection failure. 3. Token/session mismatch between systems.

v. Manage Campus Life & Events

Field	Description
Use Case ID	UC-05
Use Case Name	Manage Campus Life & Events
Actors	User
Preconditions	User is logged in
Basic Flow	1. Select Campus Life & Events 2. Display activities 3. RSVP/Register/interact
Alternative Flows	Event data unavailable → system message
Postconditions	User interacts with campus events successfully
Exceptions	1. Event registration limit exceeded. 2. Duplicate RSVP attempt. 3. Database write failure during registration.

vi. Mentorship & Alumni Connection

Field	Description
Use Case ID	UC-06
Use Case Name	Mentorship & Alumni Connection
Actors	User
Preconditions	User is logged in
Basic Flow	1. Select Mentorship & Alumni 2. Display programs/connections 3. Request mentorship/connect
Alternative Flows	Connection request fails → error message
Postconditions	User connects with mentors or alumni
Exceptions	1. Mentor no longer available. 2. Duplicate mentorship request. 3. Database failure while saving the request.

vii. Get Campus Navigation

Field	Description
Use Case ID	UC-07
Use Case Name	Get Campus Navigation
Actors	User, Visitor
Preconditions	User/Visitor can access system interface
Basic Flow	1. Select Campus Navigation 2. Connect to Campus Map API 3. Request directions 4. Display optimal route
Alternative Flows	Map API unavailable → error message
Postconditions	User/Visitor navigates campus effectively
Exceptions	1. Invalid location entered. 2. GPS/location service disabled. 3. Map API response corrupted.

viii. UC-08: Post Announcements & News

Field	Description
Use Case ID	UC-08
Use Case Name	Post Announcements & News
Actors	Admin
Preconditions	Admin logged in with proper role
Basic Flow	1. Select Post News 2. Enter announcement details 3. System posts to dashboard
Alternative Flows	Posting fails → error message
Postconditions	Announcements visible to all users
Exceptions	1. Missing required fields in announcement. 2. File attachment upload failure. 3. Unauthorized admin role.

ix. Manage User Roles

Field	Description
Use Case ID	UC-09
Use Case Name	Manage User Roles
Actors	Admin
Preconditions	Admin logged in
Basic Flow	1. Select Manage Roles 2. Assign/modify/revoke roles 3. System updates privileges
Alternative Flows	Role change fails → error message
Postconditions	User roles updated and access rights enforced
Exceptions	1. Attempt to assign invalid role. 2. Admin attempts to revoke own super-admin rights. 3. Role update conflicts with system policies.

x. Manage Emergency Contacts

Field	Description
Use Case ID	UC-10
Use Case Name	Manage Emergency Contacts
Actors	Admin
Preconditions	Admin logged in
Basic Flow	1. Select Emergency Contacts 2. Add/edit/delete contacts 3. Update database
Alternative Flows	Update fails → error message
Postconditions	Emergency contacts updated and accessible
Exceptions	1. Duplicate contact entry. 2. Invalid contact information format. 3. Database write failure.

xi. Manage Campus Location Data

Field	Description
Use Case ID	UC-11
Use Case Name	Manage Campus Location Data
Actors	Admin
Preconditions	Admin logged in
Basic Flow	1. Select Manage Location Data 2. Update building/facility data 3. Save updates
Alternative Flows	Update fails → error message
Postconditions	Campus map data accurate and current
Exceptions	1. Invalid building coordinates entered. 2. Map data format error. 3. Conflict with existing location records.

xii. Monitor System Performance

Field	Description
Use Case ID	UC-12
Use Case Name	Monitor System Performance
Actors	Admin
Preconditions	Admin logged in
Basic Flow	1. Select Monitor Performance 2. View usage, uptime, errors 3. Review and act
Alternative Flows	Metrics fail to load → error message
Postconditions	Admin has up-to-date performance information
Exceptions	1. Metrics service unavailable. 2. Incomplete monitoring data returned. 3. Unauthorized performance access.

xiii. Alumni Use Cases

Field	Description
Use Case ID	UC-13
Use Case Name	Alumni Use Cases
Actors	Alumni
Preconditions	Alumni account exists or alumni registers
Basic Flow	1. Alumni logs in 2. Register as alumni 3. Network with alumni 4. Access alumni content
Alternative Flows	Registration fails → error Networking fails → retry message
Postconditions	Alumni engages with alumni network and services
Exceptions	1. Alumni verification failure. 2. Duplicate alumni registration. 3. Networking connection request rejected.

xiv. Profile Management & Announcement Viewing

Field	Description
Use Case ID	UC-14
Use Case Name	Profile Management & Announcement Viewing
Actors	User
Preconditions	User logged in
Basic Flow	1. Edit profile information 2. View admin announcements
Alternative Flows	Update fails → error Announcements fail to load → error
Postconditions	Profile updated and announcements accessible
Exceptions	1. Invalid profile data format. 2. Profile update conflict. 3. Unauthorized profile edit attempt.

xv. Visitor Use Cases

Field	Description
Use Case ID	UC-15
Use Case Name	Visitor Use Cases
Actors	Visitor
Preconditions	Visitor can access system interface
Basic Flow	1. Get campus navigation 2. Interact with AI assistant
Alternative Flows	System unavailable → error message
Postconditions	Visitor navigates campus and accesses AI guidance
Exceptions	1. Restricted feature access attempt. 2. API request rate limit exceeded. 3. Session timeout.

4.3.4 USE CASE DIAGRAM



Figure 3: Use Case Diagram.

4.4. CONTEXT DIAGRAM

To further clarify the system boundaries, we developed a context diagram that outlined how data flows between external entities (such as users and external systems like ARIS and Google Maps API) and the UDSM Mobile Application. This helped in identifying how information will be exchanged between the system and its stakeholders.

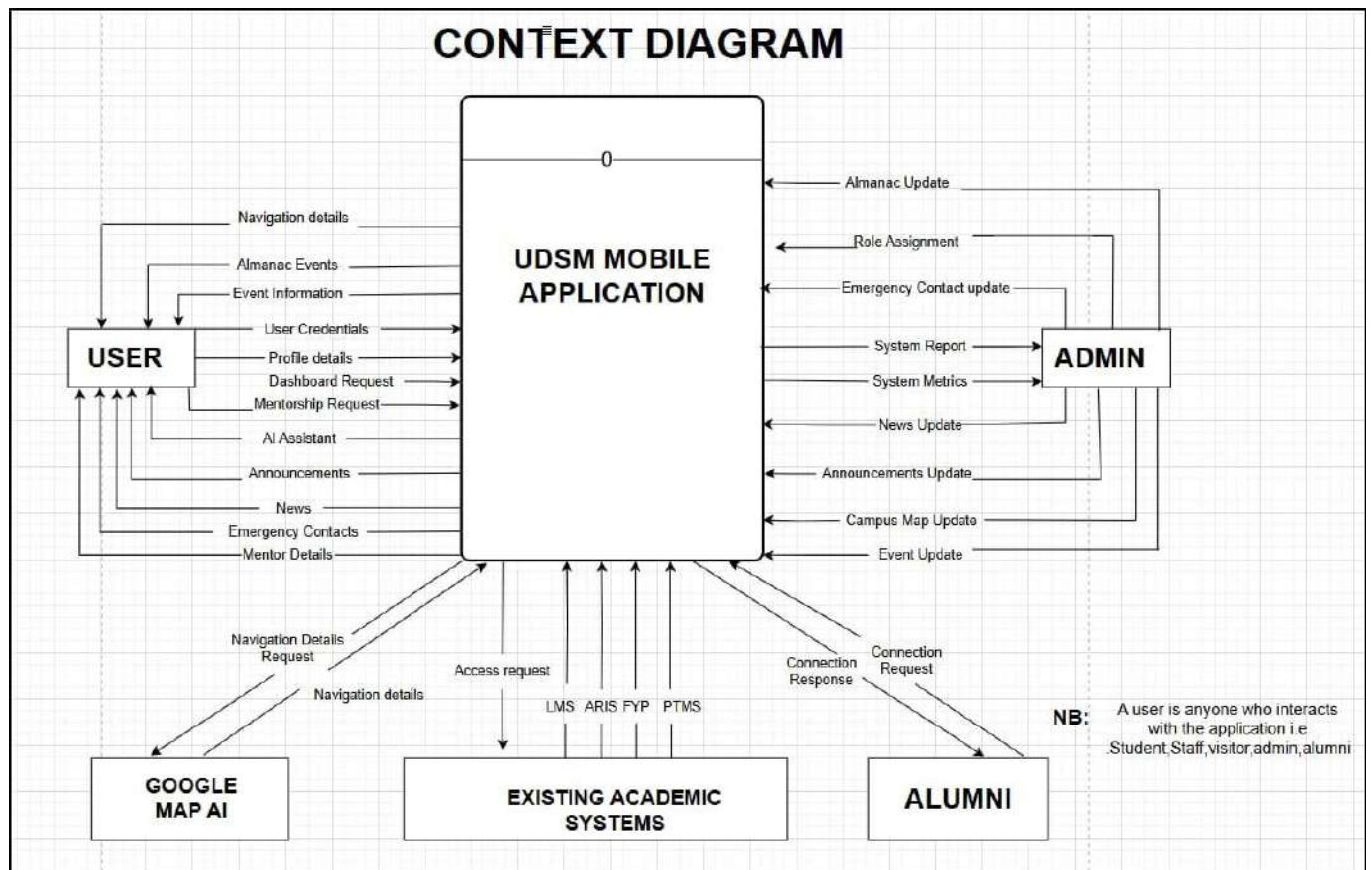


Figure 4: Context Diagram.

4.5. ENTITY RELATIONSHIP DIAGRAM AND ARCHITECTURE DIAGRAM

Since the application needs to store and manage different types of data, such as user details, announcements, news articles, campus locations, emergency contacts, and event subscriptions, a

database structure was designed to organize this data efficiently. An Entity-Relationship Diagram (ERD) was designed to show how different database tables relate to one another.

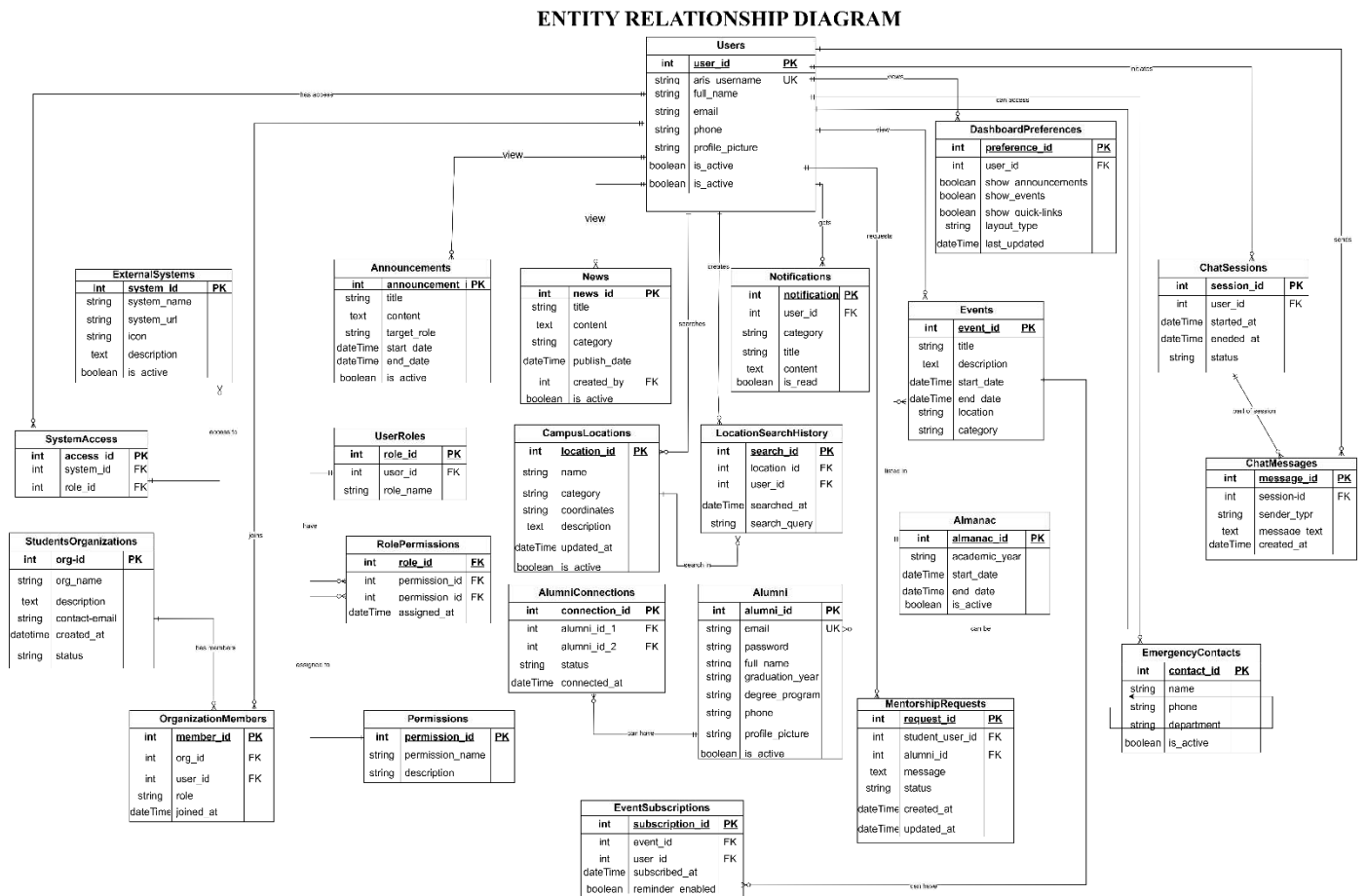
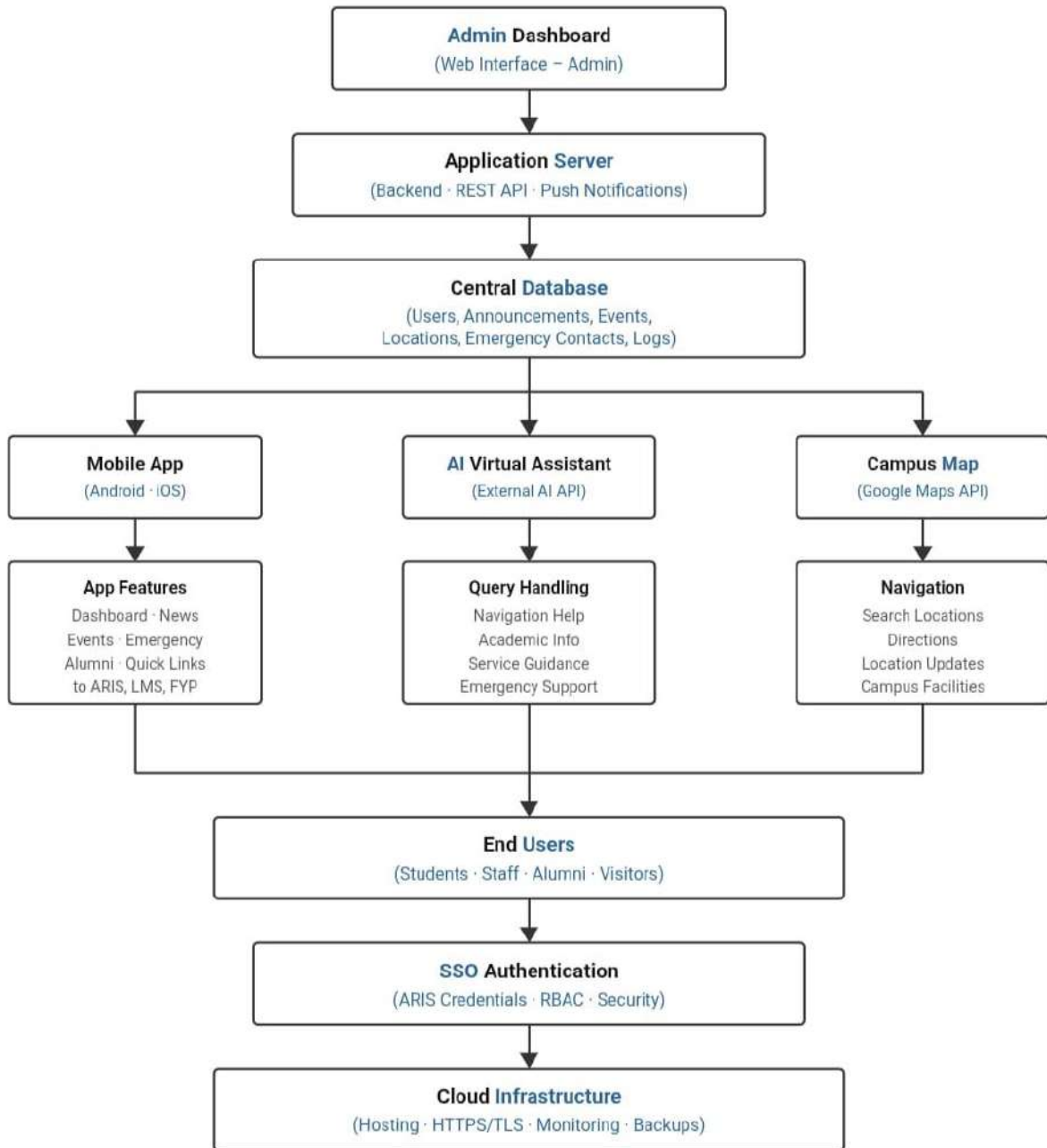


Figure 5: Entity Relationship Diagram

ARCHITECRURE DIAGRAM

HIGH LEVEL SYSTEM ARCHITECTURE



4.6. PROTOTYPE

This section shows the prototype of the system, which is a simple visual preview of how the application will look and work. It helps to demonstrate the main screens and features, such as creating an account, signing in, and navigating through the app.

The prototype gives a clear idea of the user interface and how users will interact with the system before the actual development is completed.

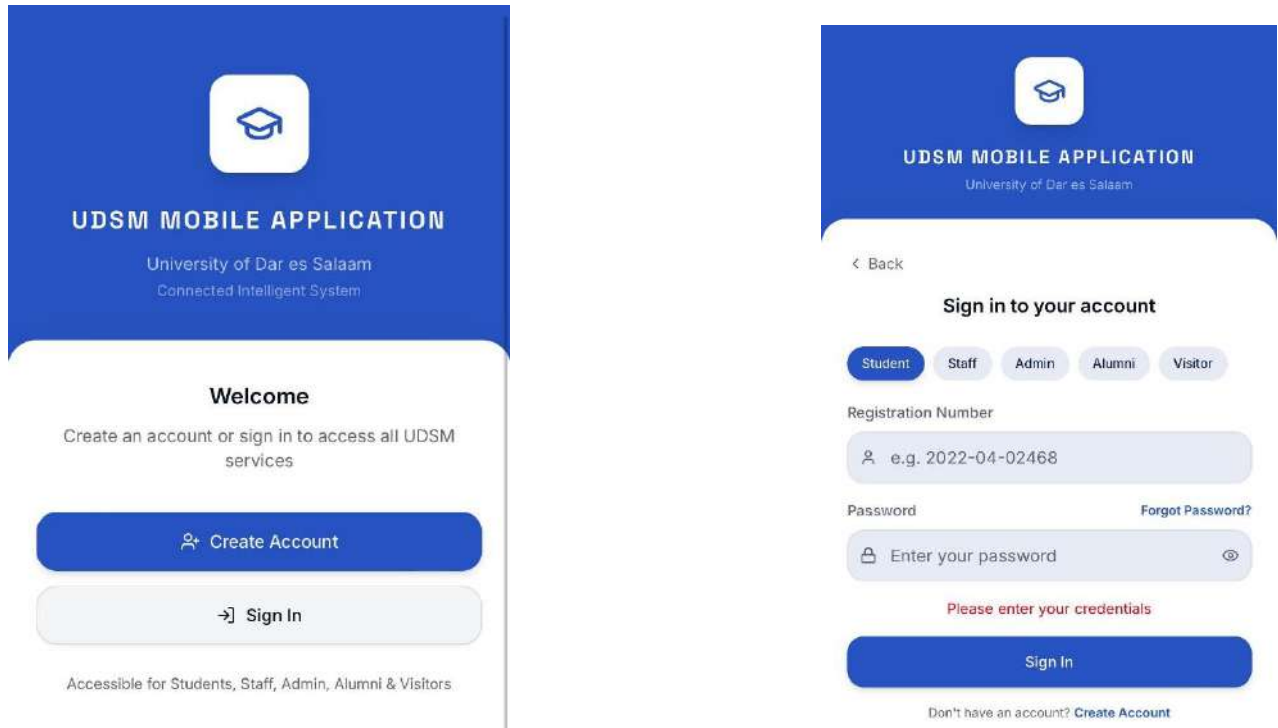


Figure 6: User Authentication Interface.

The figure above shows the prototype design of the user authentication module of the system, which includes the **Create Account** and **Sign-In** pages. These interfaces allow new users to register by providing the required credentials and enable existing users to securely access the system using their login details. This authentication feature is essential for controlling access to personalized services within the UDSM Connected Intelligent System.

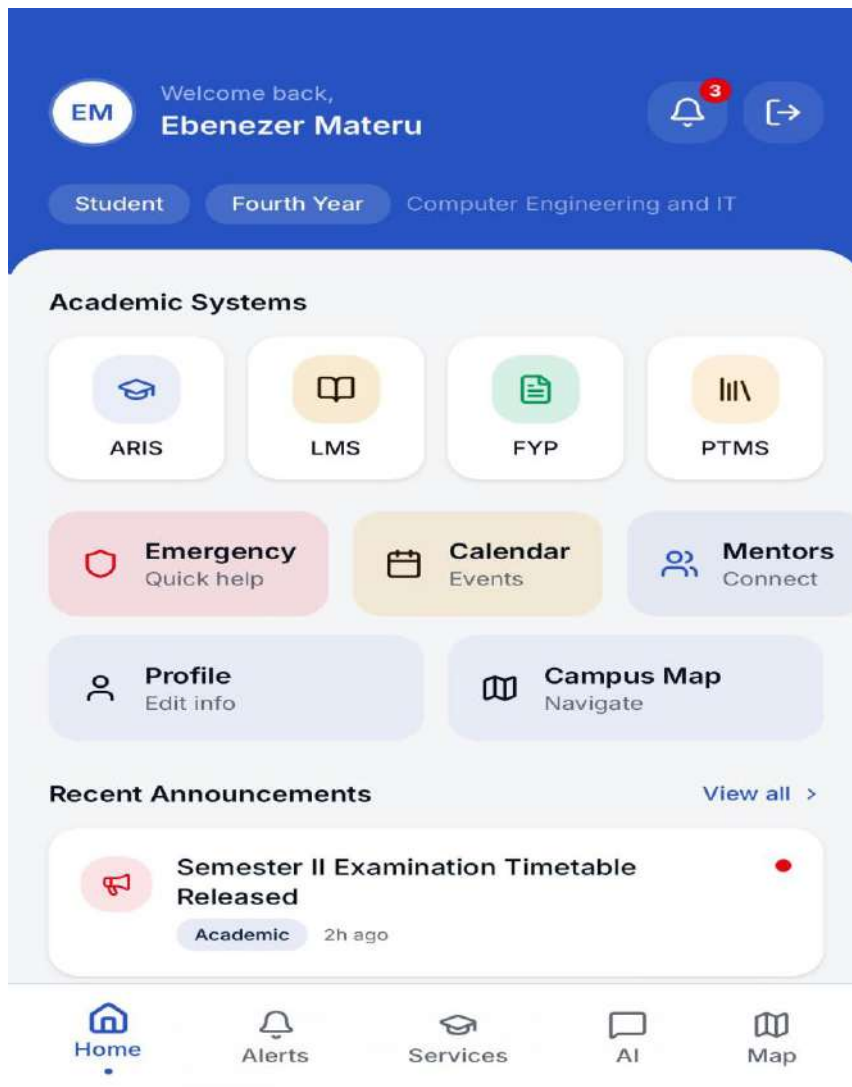


Figure 7: Student's Dashboard

This figure presents the prototype of the student interface, designed to allow students to easily access university services, updates, and system features through a user-friendly dashboard.

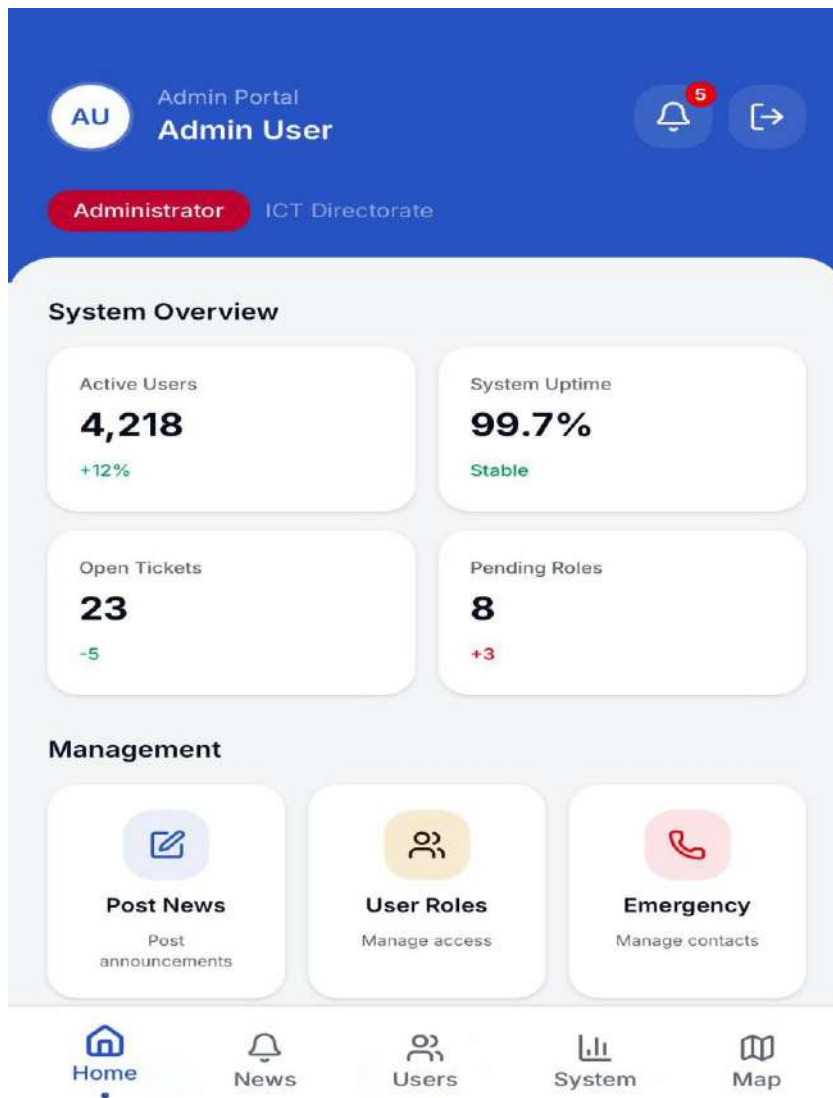


Figure 8: Admin Dashboard

This figure shows the prototype of the admin interface, which enables administrators to easily navigate through key sections of the system. Through the News section, admins can post and manage announcements and updates; in the Users section, they can view and manage user accounts and roles; the System section enables monitoring and control of system features and settings; and the Map section allows administrators to manage campus location information. This navigation structure helps administrators efficiently manage content, users, and overall system operations in a user-friendly interface.

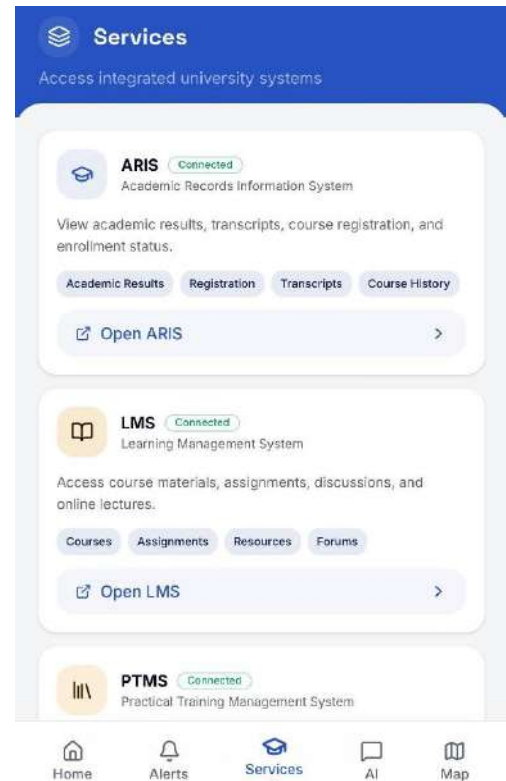
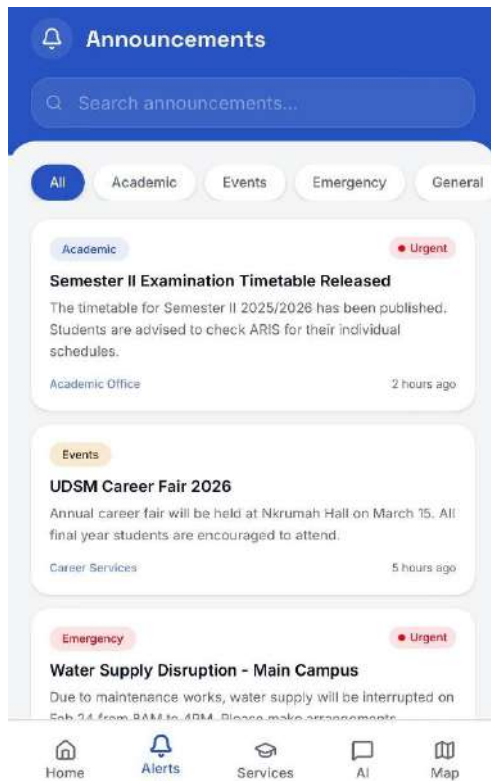


Figure 9: Alerts and Services pages

The figure shows the prototype of the Alerts and Services pages of the application. The Alerts page displays announcements and important updates to keep users informed about university information and activities. The Services page provides quick access to important university web systems such as ARIS, LMS, FYP, and PTMS through direct links, making it easier for users to navigate different platforms in one place. These pages improve accessibility and ensure that essential services and notices are centralized within the system.

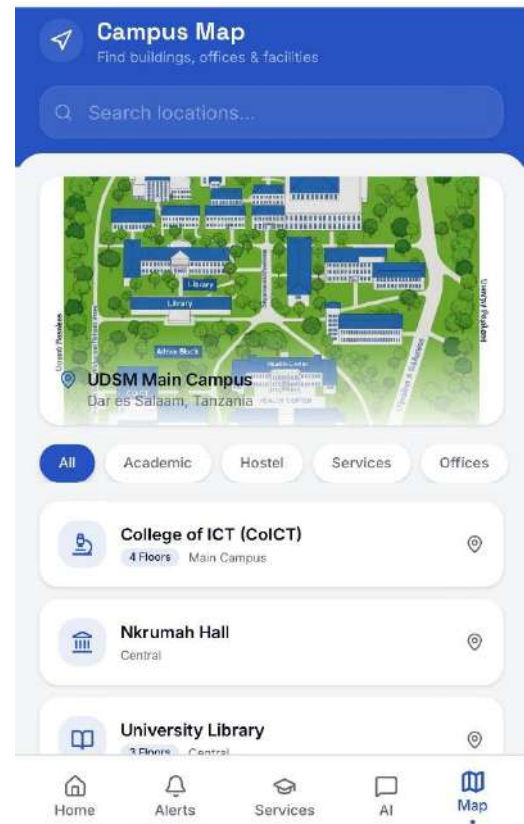
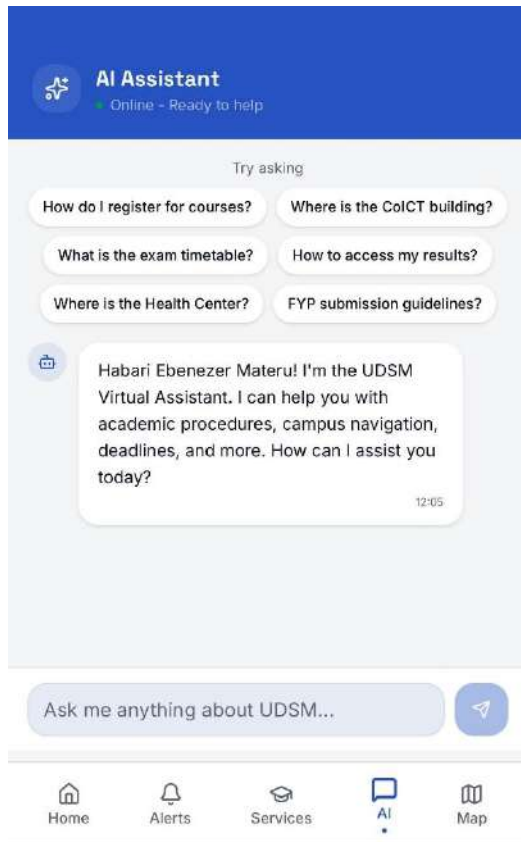


Figure 10: AI assistant and Campus Map Interfaces.

The figure shows the prototype of the AI Assistant and Campus Map interfaces of the application. The AI Assistant is designed to help users quickly get information and guidance about university services and system features, while the Campus Map allows users to easily navigate the university environment and locate important facilities.

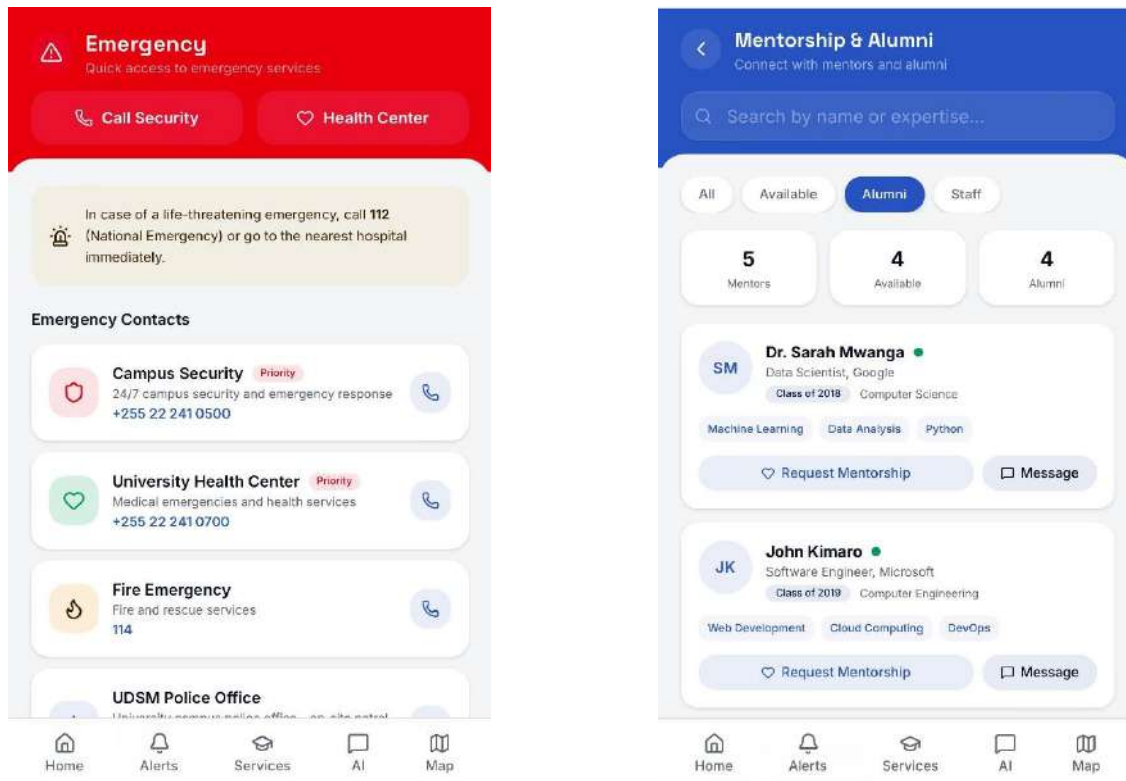


Figure 11: Emergency contacts and Mentorship&Alumni pages.

The figure shows the prototype of the Emergency Contacts, Mentorship, and Alumni pages. The Emergency Contacts page provides users with quick access to important contact information for urgent situations is the university health center and UDSM Police Office contacts. The Mentors& Alumni page is designed to connect students with mentors for guidance and academic support and engage with alumni-related information and opportunities.

4.7. CONCLUSION

To conclude, the **UDSM Connected Intelligent System** is a smart solution that makes it easier for students and staff to access various university services. Instead of searching for information in different places, everything is brought together into one easy-to-use platform. This helps improve communication and simplifies administrative tasks, ensuring everyone can quickly find important information.

The system includes features like dashboards and access to other tools, making it useful for both studying and managing university operations. With the initial prototype, we've confirmed that it works well, is easy to use, and is reliable. This project not only helps the university run more smoothly but also sets the stage for ongoing improvements and innovations in how the campus is managed.

5. REFERENCES

- Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., ... & Thomas, D. (2001). Manifesto for Agile Software Development. Retrieved from <https://agilemanifesto.org>
- Google. (2024). Google Maps API Documentation. Retrieve <https://developers.google.com/maps>
- Highsmith, J. (2009). Agile Project Management: Creating Innovative Products (2nd ed.). Addison Wesley
- University of Notre Dame. (2023). Notre Dame Mobile App Features. Retrieved from <https://oit.nd.edu>
- University of Houston. (2023). UH Go Mobile App Overview. Retrieved from <https://uh.edu/mobile>